

ELE529E Embedded Systems Project Progress Report

Course: ELE529E Embedded Systems
Submission Date: [22/05/2026]

1. Project Overview

Project Title	[Intelligent Dual-Axis Solar Tracker with Predictive Filtering]
Team Members	[Makbule Ozge Ozler] ([704252003]), [Student 2 Name] ([Student 2 No])
Project Start Date	[13/03/2026]
Expected Completion	[22/05/2026]

2. Project Milestones & Delivery Plan

Milestone	Tasks	Deadline	Status (✓/✗)
1. Requirement Analysis	Define project scope, objectives, constraints and STM32 peripheral selection.	[20/03/2026]	✓
2. System Design	Hardware/software architecture, UML diagrams, FreeRTOS task prioritization, Resource Protection (Mutex).	[10/04/2026]	✓
3. Prototype Development	Implement core functionalities (e.g., sensor integration, Adaptive EMA Filter, Custom INA219/ESP8266 Drivers, IWDG).	[30/04/2026]	✓
4. Testing & Debugging	Unit tests, system validation, performance analysis, jitter-free validation, power consumption profiling via ThingSpeak.	[15/05/2026]	⚠ (Kismen tamamlandı, ThingSpeak entegrasyonu devam ediyor)
5. Final Demo & Report	Final system integration on pertinax, demo video preparation, submit final report, prepare presentation.	[22/05/2026]	✗

3. Individual Contribution Plans

Student 1: [Makbule Özge Özler]

Task	Responsibility	Estimated Time
Software Architecture	Designing the FreeRTOS state-machine, Mutex integration, and UML logic flow.	2 weeks
Data Processing	Developing the Adaptive Exponential Moving Average (EMA) filter for noisy LDR data.	2 weeks
Cloud Integration	Managing UART communication (Circular Buffer) for ESP8266 & ThingSpeak API.	1 week
Hardware Setup	STM32F410RB configuration, sensor interfacing, and LM2596S power regulation.	2 weeks
Low-Level Drivers	Custom INA219 (I2C) and ESP8266 (UART/Interrupt) thread-safe driver development.	1.5 weeks
System Robustness	Independent Watchdog (IWDG) configuration and hardware-level performance tuning.	1 week

Student 2: [Student Name]

Task	Responsibility	Estimated Time
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4. Project Quality Utility Tree

Objective: Ensure industrial reliability, high performance, fault tolerance and usability.

1. Functional Correctness

├─ 1.1 Positioning Accuracy ($\pm 1^\circ$ error margin via Adaptive EMA Filter)

├─ 1.2 Power Monitoring (INA219 precision $< 1\%$ error)

└─ 1.3 Asynchronous Telemetry (Zero data loss via UART Circular Buffer)

2. Robustness

├─ 2.1 Resource Protection (Mutex locks on I2C and UART buses)

├─ 2.2 Fault Recovery (Independent Watchdog – IWDG auto-reset on crash)

└─ 2.2 Environmental Noise Handling (Dead-band logic to prevent servo jitter)

3. Maintainability

├─ 3.1 Modular Code (Custom drivers separated from auto-generated HAL code)

├─ 3.2 Real-Time Monitoring (Live telemetry visualization via ThingSpeak)

└─ 3.3 Documentation (Doxygen + GitHub Wiki)

5. Project Structure Diagram (PlantUML)

```

@startuml
package "Hardware Layer" {
[STM32F410RB] --> [4x LDR Sensors (ADC+DMA)]
[STM32F410RB] <--> [INA219 Power Monitor (I2C)]
[STM32F410RB] --> [2x SG90 Servos (PWM)]
[STM32F410RB] <--> [ESP8266 (UART+Interrupt)]
[STM32F410RB] --> [Independent Watchdog (IWDG)]
}

package "Software Layer (FreeRTOS)" {
[FreeRTOS Kernel] --> [DataAcqTask : ADC/I2C Reading & IWDG Refresh]
[FreeRTOS Kernel] --> [ControlTask : Adaptive EMA & PWM Control]
[FreeRTOS Kernel] --> [CommTask : UART Circular Buffer & HTTP GET]

[DataAcqTask] ..> [I2C_Mutex] : Locks
[CommTask] ..> [UART_Mutex] : Locks
}

[ESP8266 (UART+Interrupt)] <--> [ThingSpeak Cloud] : Wi-Fi / REST API
@enduml

```

6. Activity Diagram (PlantUML)

```

@startuml
start
:Initialize Peripherals (ADC-DMA, I2C, PWM, UART);
:Initialize FreeRTOS & IWDG;
fork
    partition DataAcqTask {
        repeat
            :Read 4x LDR via DMA;
            :Acquire I2C_Mutex;
            :Read Voltage/Current from INA219;
            :Release I2C_Mutex;
            :Refresh IWDG (System Alive);
            :osDelay(50ms);
        repeat while (System Running?)
    }
fork again
    partition ControlTask {
        repeat
            :Apply Adaptive EMA Filter;
            :Calculate Pan/Tilt Error;
            if (Error > Dead-band Threshold?) then (Yes)
                :Update PWM Duty Cycle (Move Servos);
            else (No)
                :Hold Position (Prevent Jitter);
            endif
            :osDelay(50ms);
        repeat
    }
end

```

```
        repeat while (System Running?)
    }
fork again
    partition CommTask {
        repeat
            :Format JSON/HTTP Payload;
            :Acquire UART_Mutex;
            :Transmit via ESP8266;
            :Release UART_Mutex;
            :osDelay(15000ms);
        repeat while (System Running?)
    }
end fork
stop
@enduml
```

7. Project Demo Setup

Hardware Requirements

- STM32F103 Development Board
- 2x SG90/MG90S Servo Motors (Pan/Tilt axes)
- 4x LDR Sensors with 10k Ω voltage dividers
- ESP8266 Wi-Fi Module
- INA219 High-Side DC Current Sensor
- LM2596S DC-DC Step-Down Converter (For isolated servo power)
- Double-sided Pertinax Board for final assembly

Software Requirements

- STM32CubeIDE (C Programming, HAL Libraries)
- FreeRTOS (CMSIS-V1/V2 RTOS integration)
- ThingSpeak Dashboard (Cloud Data Visualization)
- Serial terminal (screen / CoolTerm)

Demo Steps

1. Power on the system
2. Verify sensor data streaming (UART debug terminal).
3. Trigger real-time control task (e.g., PID response).
4. Monitor power consumption (if applicable).
5. System Startup: Power is applied. The system initializes Mutexes, IWDG, and moves servos to their calibrated home position.
6. Robust Light Tracking: A flashlight is moved across the LDRs. The Adaptive EMA filter ignores sudden shadows while smoothly tracking the primary light source without servo jitter.

7. Data Monitoring: Instantaneous current and voltage data are retrieved via thread-safe I2C and exported to the ThingSpeak Dashboard.
8. Failure Scenario (Fault Recovery): A simulated infinite loop is introduced; the IWDG resets the STM32 to demonstrate industrial-grade fault tolerance.

8. Risks & Mitigation

Risk	Probability	Impact	Mitigation Strategy	Status
Sensor Noise	Medium	High	Adaptive EMA filter + dead-band	Implemented
RTOS Stack Overflow	Low	Critical	Use <code>uxTaskGetStackHighWaterMark()</code>	Tested
DMA Cache Coherency	High	Medium	Use <code>SCB_CleanDCache()</code>	
Data Corruption on Shared Buses	High	Critical	Implemented osMutexWait/Release for thread-safe I2C and UART operations.	Implemented
Servo Jitter / Mechanical Noise	High	Medium	Developed an Adaptive EMA Filter and Dead-band logic to ignore minor variance.	Implemented
MCU Freeze / Wi-Fi Timeout	Medium	Critical	Activated Independent Watchdog (IWDG) to auto-reset the system upon failure.	Tested
Power Spikes from Servos	Medium	High	Isolated servo power lines using LM2596S regulator with common ground to STM32.	Hardware
Wi-Fi Timeout	Medium	Low	Retry logic (planned)	In progress

9. Conclusion

This report outlines the planned development of **Intelligent Dual-Axis Solar Tracker with Predictive Filtering**, including:

- ✓ Task distribution among team members.
- ✓ Key milestones with deadlines.
- ✓ Quality assurance strategies.
- ✓ UML diagrams for system design.
- ✓ Implementation of custom, thread-safe hardware drivers(INA219, ESP8266).
- ✓ Integration of FreeRTOS with Mutex resource protection.
- ✓ High system robustness via IWDG fault recovery mechanisms.
- ✓ Advanced sensor noise reduction using an Adaptive EMA algorithm.

Completed Features:

- Full FreeRTOS task structure (DataAcq, Control, Comm)
- Thread-safe INA219 driver (Open/loctl/Read/Write model)
- Thread-safe ESP8266 driver with ring buffer and interrupt callback
- Adaptive EMA filter and dead-band servo control
- IWDG refresh in all tasks (system reset on hang)
- LDR reading via ADC-DMA (4 channels)
- Servo control via TIM3 PWM (PA6, PA7)
- UART debug output (USART2 Ready!, Task02 alive, etc.)
- GitHub repository with all source code and documentation

Next Steps: Transitioning from the breadboard prototype to the final soldered pertinax board.

Conducting long-term power efficiency profiling. Completing ThingSpeak integration (ESP8266 HTTP GET – partial). Recording final demo video recording. Preparing final report write-up and poster.

Appendices

- [A] Datasheets (STM32L476RG, INA219, ESP8266, SG90)
 - [B] GitHub Repository Link
https://github.com/ozgeozler93/STM32F410_PhD_Projects/tree/main/02_Solar_Tracker_Project
 - [C] Meeting Minutes (Weekly Updates)
 - [D] Demo video (to be uploaded)
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